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EXTENDED ABSTRACT

Towards a distributional autoencoder for climate counterfactuals

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1. Introduction and objectives

A fundamental goal in climate attribution is to estimate how forced climate change contributed to certain (extreme) weather events as opposed to internal variability. In this context, the storyline attribution method compares an observed weather event, defined by its atmospheric dynamic state (i.e., atmospheric circulation), in the current ('factual') climate to the same event in a hypothetical 'counterfactual' climate [Shepherd, T. (2016)]. Hence, storyline attribution approaches aim to constrain circulation-related atmospheric variability to the observed state, which allows to focus on forced thermodynamic changes. However, simulation techniques that reconstruct the weather event in a counterfactual climate conditional on the atmospheric circulation suffer from nontrivial transferability across different climate background states. Here we demonstrate the power of distributional autoencoders (DAEs) to model the full conditional distribution of spatially resolved European temperature fields in different background climates, given the state of atmospheric circulation and the mean global warming level. This study introduces a new deep learning approach that complements the toolbox of storyline attribution.

2. Data and methods

We train, validate, and test the DAE using the Community Earth System Model 2 Large Ensemble (CESM2-LE) [Rodgers K. B.(2021)] and three additional transient CESM2 simulations. Further, we test the generated counterfactual temperature distributions against CESM2 storyline counterfactual simulations (nudged-circulation simulations [Singh J. B.(2025)]). We use 5-day mean anomalies of temperature and geopotential height fields (Z500, representing a proxy of atmospheric circulation). Furthermore, for specifying the background climate, we use the ensemble mean global mean temperature (fGMT) as a proxy of the forced response.

The DAE consists of an encoder, decoder, and a model regressing the latent space variables onto the set of covariates (Z500 and fGMT). All model components consist of stochastic neural networks [Shen X.(2024)]. During training, the model encodes and decodes fields of European surface air temperature while simultaneously predicting the latent space from the covariates. The training objective in this distributional modeling task is based on the energy score [Gneiting T.(2007)], which ensures predictions that are accurate yet exhibit variability that is consistent with the circulation conditional variability present in the training data. At time of inference, the latent space variables are predicted

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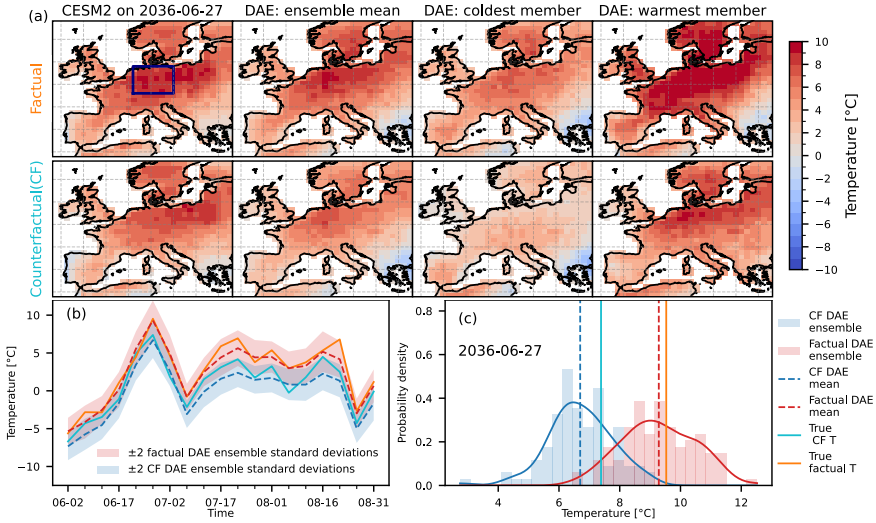

Figure 1. *DAE-generated factual and counterfactual (CF) temperature distributions for an exemplary 5-day event from the CESM2 test simulation.* ***(a)*** *Factual and counterfactual events simulated by CESM2 (first column), with the remaining columns showing the corresponding DAE-generated ensemble.* ***(b)*** *Time series of summer 2036 temperatures averaged over the domain box shown in (a).* ***(c)*** *Empirical factual and counterfactual distributions of DAE-generated, domain-averaged temperatures, with Gaussian kernel density estimates of the corresponding probability density functions.*

from Z500 and fGMT, and subsequently decoded into 100 ensemble members each representing a temperature field and thereby sampling the desired distributions.

3. Preliminary results and discussion

We perform a storyline attribution case study of summer temperatures averaged over a central European domain, focusing on a heat event around June 27, 2036. The event is translated into a counterfactual background climate corresponding to no forced global warming, as illustrated in Figure 1. The generated DAE ensembles clearly illustrate the difference in mean intensity and probability for such temperatures to occur under the given atmospheric circulation. Evaluated over the entire test set and in comparison to a baseline linear quantile regression model, the DAE shows overall very promising performance (not shown). In line with storyline attribution, our approach highly constrains factual and counterfactual temperatures to conditional distributions that are plausible to occur under the given atmospheric circulation. This can ultimately improve the approximation of uncertainties in storyline attribution. Finally, we expect our approach to perform well in transfer applications to reanalysis data.

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Data and code availability. The data used in this abstract is available at https://doi.org/10.5281/zenodo.19356777, and the code for the analysis can be accessed at https://doi.org/10.5281/zenodo.19360130.

References

[Shepherd, T. (2016)] Shepherd, T. (2016). A Common Framework for Approaches to Extreme Event Attribution. *Current Climate Change Reports* **2, No. 1**, 28-38.

[Rodgers K. B.(2021)] Rodgers K. B.(2021). Ubiquity of human-induced changes in climate variability. *Earth System Dynamics* **12, No. 4**, 1393-1411.

[Shen X.(2024)] Shen X.(2024). Engression: Extrapolation through the Lens of Distributional Regression. *arXiv*

[Singh J. B.(2025)] Singh J. B.(2025). Externally forced circulation changes amplify mid-latitude regional heat extremes in climate model nudged-circulation experiments. *Research Square* **Preprint, v. 2**.

[Gneiting T.(2007)] Gneiting T.(2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association* **102 No. 477**, 359–378.